

Do NFL Preseason Wins Predict Regular-Season Success?

A Longitudinal Linear Regression Study of 798 Team-Seasons

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Abstract

This study evaluates whether National Football League (NFL) preseason performance provides useful predictive information about regular-season success. A team-season dataset was constructed for 2000–2025, excluding 2020 because the preseason was canceled, yielding 798 historical observations. Preseason features included win percentage, points per game, points allowed per game, and point differential per game. Regular-season win percentage was used as the response variable to normalize the change from a 16-game to a 17-game regular season. Models were evaluated using a chronological split: 2000–2019 for training ($n = 638$) and 2021–2025 for testing ($n = 160$). Although preseason point differential and preseason win percentage showed weak positive historical correlations with regular-season win percentage ($r = 0.155$ and $r = 0.105$, respectively), neither translated into useful out-of-sample predictive performance. A baseline model predicting the training-set mean achieved a test MAE of 0.1531, whereas the best regression model, using preseason win percentage alone, achieved an MAE of 0.1539 and $R^2 = -0.0279$. More complex specifications performed worse. A descriptive refit on all historical observations explained only 1.1% of the variance in regular-season win percentage. The results indicate that preseason win-loss outcomes contain little practical forecasting value for regular-season success under simple linear models, and they illustrate the importance of chronological validation and baseline comparison in sports machine-learning studies.

Keywords: linear regression; sports analytics; NFL; machine learning; out-of-sample validation; preseason; predictive modeling

1 Introduction

Linear regression is one of the foundational methods in supervised statistical learning because it provides an interpretable relationship between one or more predictors and a quantitative response [1]. In sports analytics, however, predictive relationships can be difficult to separate from noise, particularly when the observed data are generated under strategic conditions that differ from the competitive environment of interest.

NFL preseason games provide a useful test case. They are real games with observable outcomes, but teams use them for roster evaluation, scheme testing, player development, and workload management rather than solely for maximizing wins. This raises a natural empirical question: *does preseason performance contain enough signal to predict regular-season success?*

The initial hypothesis of this project was that teams with stronger preseason results—especially higher win percentages and larger scoring margins—might exhibit higher regular-season win percentages. The study was designed not merely to fit a line to historical data, but to evaluate whether any apparent relationship generalizes to later, unseen seasons. This distinction between fit and prediction is central to statistical learning [1].

The contributions of this study are threefold. First, it constructs a reproducible team-season dataset spanning 25 usable preseasons from 2000 through 2025. Second, it compares simple and multiple linear-regression models against a naive average-team baseline using a chronological holdout period. Third, it freezes a 2026 preseason-only prediction set to demonstrate how an apparently intuitive sports hypothesis can fail under out-of-sample evaluation.

2 Data

2.1 Sources and observation unit

The unit of analysis is one NFL team in one season. Preseason game schedules and scores were obtained from ESPN’s public NFL team schedule endpoint using preseason season type queries. Regular-season and postseason game data were obtained through the nflverse ecosystem using `nflreadpy`. Historical franchise codes were normalized so that relocations and rebrandings mapped to a consistent contemporary team identifier (for example, OAK $\rightarrow$ LV, SD $\rightarrow$ LAC, STL/LA $\rightarrow$ LAR, and WSH $\rightarrow$ WAS).

The historical modeling sample covers 2000–2019 and 2021–2025. The 2020 season was excluded because there was no NFL preseason. The final historical sample contains 798 team-seasons: 31 teams in each of 2000 and 2001, and 32 teams in each usable season from 2002 onward. An additional 32 rows contain 2026 preseason features only; all 2026 regular-season target values were intentionally left blank to prevent leakage.

2.2 Feature construction

For team i in season s , preseason win percentage was defined as

$$\text{PreWinPct}_{is} = \frac{W_{is} + 0.5T_{is}}{G_{is}}, \quad (1)$$

where W , T , and G denote preseason wins, ties, and games played. Scoring features were normalized per game to account for differences in preseason schedule length:

$$\text{PrePPG}_{is} = \frac{PF_{is}}{G_{is}}, \quad (2)$$

$$\text{PrePAPG}_{is} = \frac{PA_{is}}{G_{is}}, \quad (3)$$

$$\text{PreDiffPG}_{is} = \frac{PF_{is} - PA_{is}}{G_{is}}. \quad (4)$$

The response variable was regular-season win percentage rather than raw wins:

$$Y_{is} = \text{RegWinPct}_{is} = \frac{W_{is}^{(r)} + 0.5T_{is}^{(r)}}{G_{is}^{(r)}}. \quad (5)$$

This choice makes seasons before and after the NFL’s transition from 16 to 17 regular-season games more directly comparable.

Table 1: Variables used in the analysis.

Variable	Definition
<code>pre_win_pct</code>	Preseason win percentage, counting a tie as one-half win.
<code>pre_ppg</code>	Preseason points scored per game.
<code>pre_pa_pg</code>	Preseason points allowed per game.
<code>pre_point_diff_pg</code>	Preseason scoring differential per game.
<code>reg_win_pct</code>	Regular-season win percentage; primary regression target.
<code>made_playoffs</code>	Binary indicator for postseason qualification; retained for future classification work.
<code>super_bowl_champion</code>	Binary indicator for Super Bowl champion; retained for future classification work.

2.3 Integrity checks

The assembled data were checked for duplicate team-season keys, inconsistent win/loss/tie totals, missing historical targets, and team-count anomalies. The final merged dataset contained zero duplicate team-seasons and zero missing historical regular-season targets. The 2026 target columns remained missing by design. As a spot check, the 2000 Baltimore Ravens preseason record aggregated to 4–0 with 74 points scored and 43 allowed; the corresponding regular season was 12–4 and culminated in a Super Bowl championship.

3 Exploratory Analysis

The initial scatterplot (Figure 1) shows substantial dispersion at every preseason win percentage. Vertical bands are expected because teams play only a small number of preseason games, so win percentage takes on a limited set of discrete values. Visually, the relationship between preseason win percentage and regular-season success is weak.

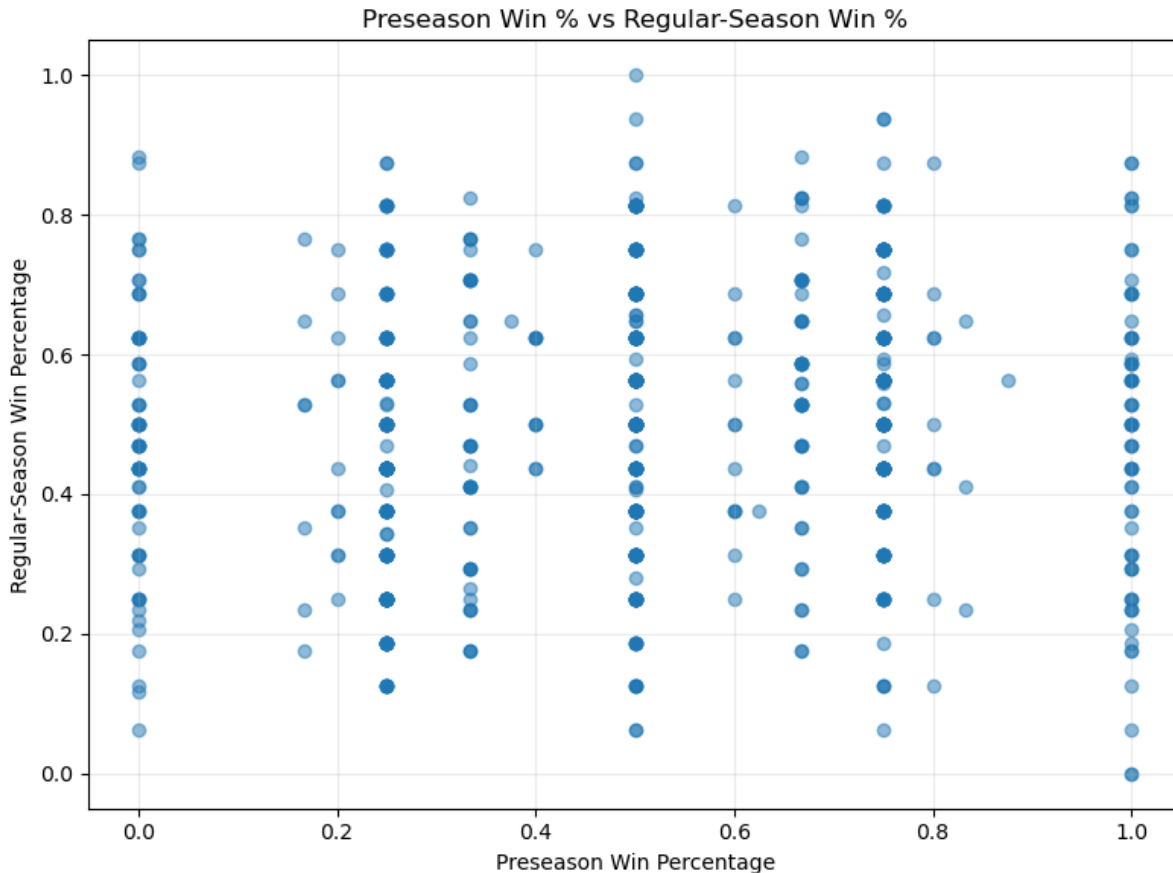


Figure 1: Historical preseason win percentage versus regular-season win percentage for 798 team-seasons. The broad vertical dispersion indicates that identical preseason records are followed by a wide range of regular-season outcomes.

Pearson correlations reinforced the visual result. Preseason point differential per game had the largest correlation with regular-season win percentage ($r = 0.155$), followed closely by preseason points per game ($r = 0.153$). Preseason win percentage correlated only $r = 0.105$ with the target, while points allowed per game correlated $r = -0.077$.

Table 2: Historical correlation with regular-season win percentage.

Preseason feature	Correlation
Point differential per game	0.155
Points per game	0.153
Win percentage	0.105
Points allowed per game	-0.077

The predictors themselves were not independent. Preseason win percentage correlated 0.807 with preseason point differential per game. Point differential also correlated 0.738 with points per game and -0.739 with points allowed per game. Because

$$\text{PreDiffPG} = \text{PrePPG} - \text{PrePAPG}, \quad (6)$$

strong dependence among these features is structurally expected. This matters for coefficient interpretation in multiple regression because multicollinearity can make individual coefficient estimates unstable even when predictive performance is the primary goal [1].

4 Methods

4.1 Linear regression

For a single predictor x_i , the simple linear-regression model is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad (7)$$

with prediction

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i. \quad (8)$$

Ordinary least squares estimates the coefficients by minimizing the residual sum of squares

$$\text{RSS} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2. \quad (9)$$

For the multiple model,

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{PreWinPct}_i + \hat{\beta}_2 \text{PreDiffPG}_i. \quad (10)$$

All regressions were fit with scikit-learn’s `LinearRegression` implementation [2].

4.2 Chronological validation

A random train/test split could allow later-era team-seasons to help explain earlier ones and would not mimic the intended forecasting use case. Therefore, the sample was divided chronologically:

- **Training:** 2000–2019, $n = 638$ team-seasons.
- **Testing:** 2021–2025, $n = 160$ team-seasons.

The 2020 season was absent from both sets because no preseason occurred.

4.3 Models and baseline

Three regression specifications were evaluated:

$$\begin{aligned} M_1 : & \text{RegWinPct} \sim \text{PreWinPct}, \\ M_2 : & \text{RegWinPct} \sim \text{PreDiffPG}, \\ M_3 : & \text{RegWinPct} \sim \text{PreWinPct} + \text{PreDiffPG}. \end{aligned}$$

A naive baseline predicted every test team using the mean training-set regular-season win percentage, which was 0.5000.

Performance was measured using mean absolute error (MAE), mean squared error (MSE), and R^2 :

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i|, \quad (11)$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2, \quad (12)$$

$$R^2 = 1 - \frac{\sum_i (Y_i - \hat{Y}_i)^2}{\sum_i (Y_i - \bar{Y})^2}. \quad (13)$$

For interpretability, MAE in win percentage was multiplied by 17 to approximate regular-season win error in the modern schedule.

5 Results

5.1 Out-of-sample model comparison

Table 3 presents held-out performance. The baseline was the best-performing approach. Predicting every team at approximately .500 produced $\text{MAE} = 0.1531$, corresponding to about 2.60 wins in a 17-game season. The best regression, M_1 , was slightly worse with $\text{MAE} = 0.1539$ and $R^2 = -0.0279$. Models using point differential performed worse still.

Table 3: Held-out performance on 2021–2025. Lower MAE/MSE is better; higher R^2 is better.

Model	R^2	MAE	MSE	Approx. wins MAE
Baseline (mean)	-0.0000	0.1531	0.0344	2.60
M_1 : Preseason win %	-0.0279	0.1539	0.0354	2.62
M_2 : Point diff./game	-0.1043	0.1602	0.0380	2.72
M_3 : Win % + point diff.	-0.1275	0.1617	0.0388	2.75

The training-period equation for M_1 was

$$\widehat{\text{RegWinPct}} = 0.4496 + 0.1008 \text{PreWinPct}. \quad (14)$$

This coefficient implies that moving from a completely winless preseason to an undefeated preseason changes predicted regular-season win percentage by only about 0.101, or roughly 1.7 wins over a 17-game season.

For M_2 , the fitted equation was

$$\widehat{\text{RegWinPct}} = 0.5000 + 0.0063 \text{PreDiffPG}. \quad (15)$$

A 10-point increase in preseason point differential per game therefore changed the predicted regular-season win percentage by only 0.063, approximately 1.1 wins over 17 games.

The multiple model M_3 produced coefficients of -0.0881 for preseason win percentage and 0.0091 for point differential per game, with intercept 0.5440. The sign reversal on the win-percentage coefficient should not be interpreted causally; the two predictors are strongly correlated ($r = 0.807$), making individual coefficients unstable. More importantly, M_3 had the worst held-out MAE and R^2 among the tested specifications.

5.2 Descriptive all-history fit

To visualize the overall relationship rather than to estimate holdout performance, the simple preseason-win model was refit on all 798 historical observations. The resulting equation was

$$\widehat{\text{RegWinPct}} = 0.463 + 0.074 \text{PreWinPct}, \quad (16)$$

with historical $R^2 = 0.011$. Thus, preseason win percentage explained only about 1.1% of the observed variation in regular-season win percentage. Figure 2 shows the shallow least-squares slope and selected residuals.

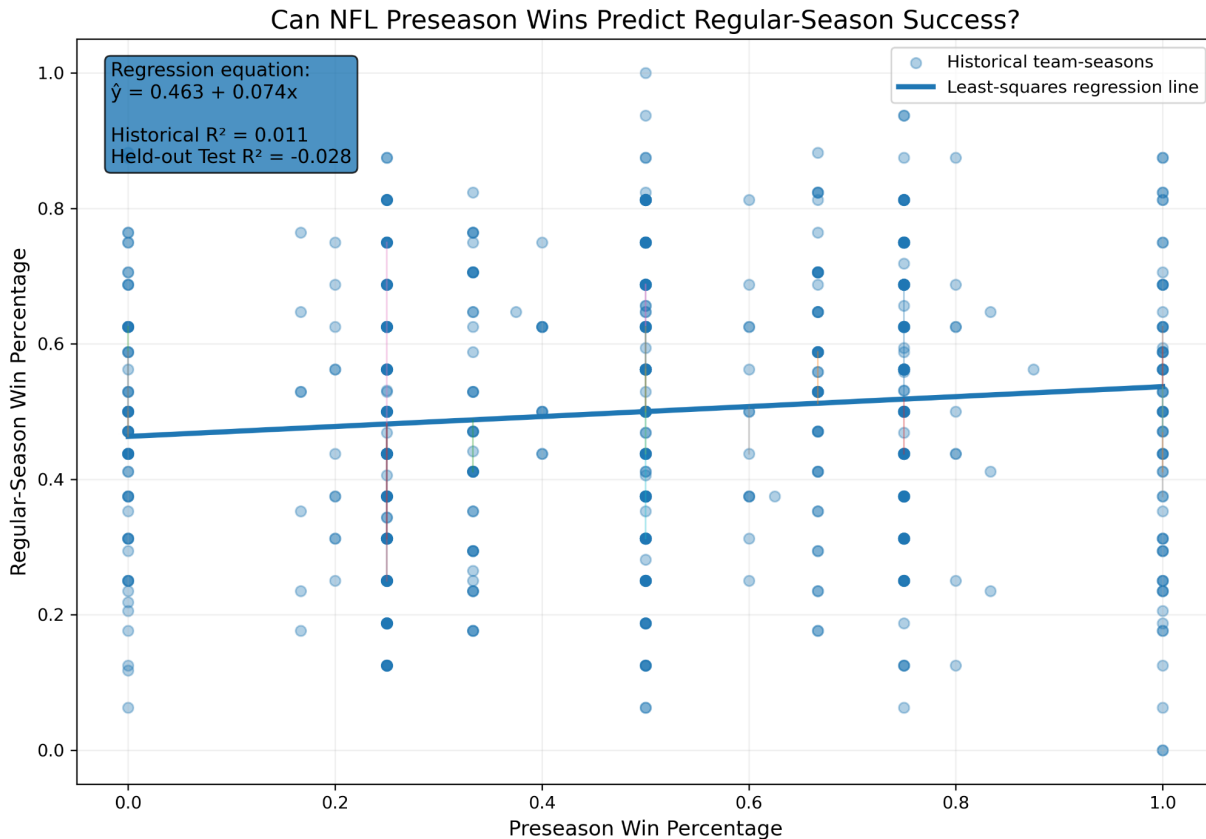


Figure 2: Least-squares fit using all 798 historical team-seasons. Selected vertical segments illustrate residuals, $e_i = Y_i - \hat{Y}_i$. The regression line is nearly flat relative to the dispersion of observed outcomes.

5.3 2026 preseason-only inference

Because M_1 was the best regression model by held-out MAE, it was retrained on all 798 historical observations and applied to 32 teams using 2026 preseason features only. The model's top tier was a four-way tie: the Los Angeles Rams, Baltimore Ravens, Buffalo Bills, and Cincinnati Bengals each had a 1.000 preseason win percentage and therefore received identical predictions of approximately 0.5368 regular-season win percentage, or 9.1 wins over 17 games.

This tie is not an artifact of ranking software; it follows directly from the model. Because M_1 uses only preseason win percentage, teams with the same input receive the same prediction. The narrow spread of predictions around .500 is itself informative: the fitted model learned that preseason record provides little leverage for distinguishing future team quality.

6 Discussion

The primary result is not that preseason performance is unrelated to regular-season success in an absolute sense. Weak positive historical associations were observed. Rather, the result is that these associations were too small and unstable to improve forecasting on later seasons under the tested linear models.

This distinction matters. If model quality were judged only by the direction of the historical slope, the study could misleadingly conclude that undefeated preseason teams should be favored. Chronological validation produced a different conclusion: all three regression models yielded negative test R^2 , and the naive average-team baseline achieved the lowest MAE and MSE. In practical terms, preseason win-loss outcomes did not add sufficient information to beat the prediction “every team is roughly average.”

The experiment also illustrates why predictive studies should compare against simple baselines. A model can produce sensible coefficients, visually plausible lines, and positive in-sample correlation while still failing to improve real forecasting performance. The negative held-out R^2 values here indicate that, under squared-error loss, the regressions were worse than using a constant prediction on the unseen period.

A second lesson concerns feature construction. Point differential appeared more correlated with regular-season success than raw preseason win percentage, but it did not generalize better. In addition, win percentage, points scored, points allowed, and point differential are mechanically and behaviorally related. Adding correlated features without a clear validation gain increased model complexity while reducing performance.

7 Limitations

Several limitations constrain interpretation.

1. **Preseason objectives differ from regular-season objectives.** Teams frequently prioritize player evaluation, health, depth-chart decisions, and scheme testing over winning.
2. **Small preseason samples.** Most teams play only three or four preseason games, making win percentage coarse and highly variable.
3. **Roster mismatch.** Preseason snaps are distributed among starters, backups, rookies, and players who may not remain on the final roster.
4. **Model simplicity.** The study intentionally focuses on linear models. Nonlinear interactions or richer football features were not modeled.
5. **Temporal dependence.** Team-seasons are not perfectly independent across years; franchises retain coaches, players, organizational structure, and strategic tendencies.
6. **Schedule and context omitted.** Strength of opponent, injuries, quarterback usage, betting-market expectations, travel, and roster quality were not included.
7. **No causal interpretation.** The regressions estimate association and predictive value, not the causal effect of winning preseason games.

8 Future Work

A natural extension is classification rather than regression. Playoff qualification can be modeled as a binary response with logistic regression, while direct Super Bowl prediction is substantially more imbalanced because there is only one champion per season. Any such study should preserve chronological validation and use metrics suitable for class imbalance, such as precision-recall analysis, rather than relying on accuracy alone.

Further work could also incorporate richer preseason features such as offensive and defensive efficiency, quarterback usage, turnovers, sacks, yards per play, roster continuity, and market-implied expectations. Rolling-origin evaluation would better quantify how predictive relationships change across NFL eras. Regularization methods could help when adding correlated predictors, but any added complexity should be justified by improved out-of-sample performance rather than in-sample fit.

9 Conclusion

Across 798 NFL team-seasons, preseason performance showed only weak historical association with regular-season success. Preseason win percentage explained approximately 1.1% of historical variance when fit across the full sample, and models based on preseason win percentage or point differential failed to outperform a simple average-team baseline on a chronological 2021–2025 holdout set. The best regression model produced a test R^2 of -0.0279 and an average error of about 2.62 wins in a 17-game season.

The central empirical conclusion is therefore straightforward: **NFL preseason win-loss performance is not a useful standalone predictor of regular-season success under simple linear regression.** The broader machine-learning conclusion is equally important: a positive historical relationship is not sufficient evidence of predictive value. Baselines, out-of-sample testing, and careful treatment of correlated features are necessary to distinguish real forecasting signal from patterns that do not generalize.

Reproducibility Notes

The analysis was implemented in Python using `pandas`, `matplotlib`, `scikit-learn`, `requests`, and `nflreadpy`. The workflow consisted of six scripts: (1) download nflverse game data, (2) download and aggregate preseason game data, (3) build the merged team-season dataset, (4) perform exploratory analysis, (5) compare linear-regression specifications and the baseline, and (6) produce the final regression visualization. The final dataset contains 830 rows total: 798 historical observations and 32 2026 preseason-only rows.

References

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